# Equilibrium balances components

**LEARNING TARGET** I can find a missing force when acceleration is zero.

## A · KEY VOCABULARY

| | |
|---|---|
| **Equilibrium** | Zero $a$: forces sum to zero on each axis. |
| **Balance** | Vertical parts support the weight. |
| **Symmetry** | Equal angles and loading: equal tensions. |
| **Rest** | $v = 0$ alone does not prove balance. |

## B · KEY EQUATIONS + UNITS

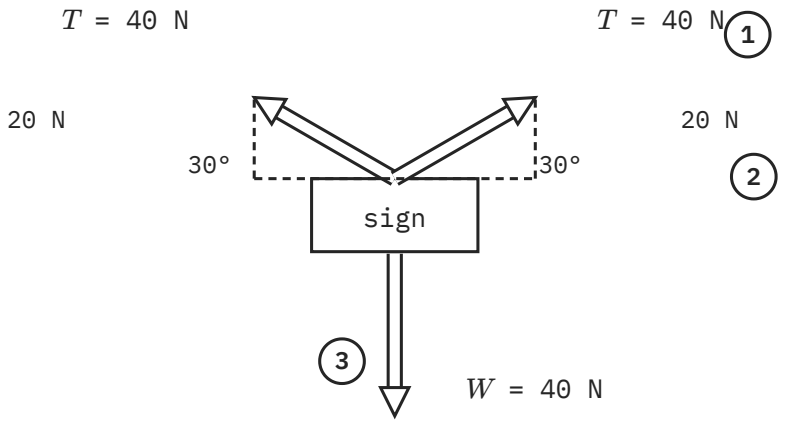
$$\Sigma F_x = \Sigma F_y = 0$$

$$2T \sin\theta = W$$

$T$ force per rope, N
$\theta$ above horizontal

## C · COLOR IT · VERTICAL PARTS ADD

*40 N sign; equal 30° ropes.*




☐ **1 · ORANGE** Rope tensions

☐ **2 · ORANGE** Vertical parts: **20 N**

☐ **3 · ORANGE** Weight: **40 N**

## D · WORKED EXAMPLE

A **40 N** sign hangs at rest from two equal ropes, each **30°** above horizontal. **Find the tension in each rope.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $W$ = **40 N**; $\theta$ = **30°**<br>equal tensions<br>$a$ = 0 | $x$ parts cancel.<br>$2T \sin$ **30°** = **40 N**. | $2T(0.5)$ = **40 N**<br>$T$ = **40 N** | **40 N per rope**<br>Each rope gives **20 N** upward. |

## E · THE 2026-XJ THREAD

DAY 18 OF 22 · NEW TOOL: COMPONENT BALANCE

Zero net force would give XJ a constant velocity, not a guaranteed stop. Analysts must check each coordinate direction before treating its motion as steady.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else